

IIT JAM 2005 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

15

TOTAL MARKS

90 M

TEST DURATION

60 Min

PAPER PATTERN

MCQ

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2005 • Q1 • ID: JAM_2005_Q1)

MCQ

+6 M

Neg: -2.0

1. Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers such that $b_n = a_{2n}$ and $c_n = a_{2n+1}$. Then $\{a_n\}$ is convergent
- (A) implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent
- (B) implies $\{c_n\}$ is convergent but $\{b_n\}$ need not be convergent
- (C) implies both $\{b_n\}$ and $\{c_n\}$ are convergent
- (D) if both $\{b_n\}$ and $\{c_n\}$ are convergent

Question 2 (JAM 2005 • Q2 • ID: JAM_2005_Q2)

MCQ

+6 M

Neg: -2.0

2. An integrating factor of $x \frac{dy}{dx} + (3x + 1)y = xe^{-2x}$ is
- (A) xe^{3x}
- (B) $3xe^x$
- (C) xe^x
- (D) x^3e^x

Question 3 (JAM 2005 • Q3 • ID: JAM_2005_Q3)

MCQ

+6 M

Neg: -2.0

3. The general solution of $x^2 \frac{d^2y}{dx^2} - 5x \frac{dy}{dx} + 9y = 0$ is
- (A) $(c_1 + c_2x)e^{3x}$
- (B) $(c_1 + c_2 \ln x)x^3$
- (C) $(c_1 + c_2x)x^3$
- (D) $(c_1 + c_2 \ln x)e^{x^3}$
- (Here c_1 and c_2 are arbitrary constants.)

Question 4 (JAM 2005 • Q4 • ID: JAM_2005_Q4)

MCQ

+6 M

Neg: -2.0

4. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. If $\phi(x, y, z)$ is a solution of the Laplace equation then the vector field $(\vec{\nabla}\phi + \vec{r})$ is
- (A) neither solenoidal nor irrotational
 - (B) solenoidal but not irrotational
 - (C) both solenoidal and irrotational
 - (D) irrotational but not solenoidal

Question 5 (JAM 2005 • Q5 • ID: JAM_2005_Q5)

MCQ

+6 M

Neg: -2.0

5. Let $\vec{F} = x\hat{i} + 2y\hat{j} + 3z\hat{k}$, S be the surface of the sphere $x^2 + y^2 + z^2 = 1$ and $\hat{n}$ be the inward unit normal vector to S . Then $\iint_S \vec{F} \cdot \hat{n} dS$ is equal to
- (A) 4π
 - (B) -4π
 - (C) 8π
 - (D) -8π

Question 6 (JAM 2005 • Q6 • ID: JAM_2005_Q6)

MCQ

+6 M

Neg: -2.0

6. Let A be a 3×3 matrix with eigenvalues 1, -1 and 3. Then
- (A) $A^2 + A$ is non-singular
 - (B) $A^2 - A$ is non-singular
 - (C) $A^2 + 3A$ is non-singular
 - (D) $A^2 - 3A$ is non-singular

Question 7 (JAM 2005 • Q7 • ID: JAM_2005_Q7)

MCQ

+6 M

Neg: -2.0

7. Let $T : \mathbf{R}^3 \longrightarrow \mathbf{R}^3$ be a linear transformation and I be the identity transformation of $\mathbf{R}^3$. If there is a scalar c and a non-zero vector $x \in \mathbf{R}^3$ such that $T(x) = cx$, then $\text{rank}(T - cI)$
- (A) cannot be 0
 - (B) cannot be 1
 - (C) cannot be 2
 - (D) cannot be 3

Question 8 (JAM 2005 • Q8 • ID: JAM_2005_Q8)

MCQ

+6 M

Neg: -2.0

8. In the group $\{1, 2, \dots, 16\}$ under the operation of multiplication modulo 17, the order of the element 3 is
- (A) 4
 - (B) 8
 - (C) 12
 - (D) 16

Question 9 (JAM 2005 • Q9 • ID: JAM_2005_Q9)

MCQ

+6 M

Neg: -2.0

9. A ring R has maximal ideals
- (A) if R is infinite
 - (B) if R is finite
 - (C) if R is finite with at least 2 elements
 - (D) only if R is finite

Question 10 (JAM 2005 • Q10 • ID: JAM_2005_Q10)

MCQ

+6 M

Neg: -2.0

10. The integral $\int_0^1 \left[\int_0^{1-z} \left(\int_0^2 dx \right) dy \right] dz$ is equal to

(A) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^2 dx \right) dz \right] dy$

(B) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^1 dx \right) dz \right] dy$

(C) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-z} dx \right) dz \right] dy$

(D) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-y} dx \right) dz \right] dy$

Question 11 (JAM 2005 • Q11 • ID: JAM_2005_Q11)

MCQ

+6 M

Neg: -2.0

11. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be continuous and $g, h: \mathbf{R}^2 \rightarrow \mathbf{R}$ be differentiable. Let $F(u, v) = \int_v^u f(t) dt$,

where $u = g(x, y)$ and $v = h(x, y)$. Then $\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} =$

(A) $f(g(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(h(x, y)) \left[\frac{\partial h}{\partial x} + \frac{\partial h}{\partial y} \right]$

(B) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

(C) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] + f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

(D) $f(g(x, y)) \left[\frac{\partial g}{\partial x} - \frac{\partial g}{\partial y} \right] + f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

Question 12 (JAM 2005 • Q12 • ID: JAM_2005_Q12)

MCQ

+6 M

Neg: -2.0

12. Let $y = f(x)$ be a smooth curve such that $0 < f(x) < K$ for all $x \in [a, b]$. Let

L = length of the curve between $x = a$ and $x = b$

A = area bounded by the curve, x -axis, and the lines $x = a$ and $x = b$

S = area of the surface generated by revolving the curve about x -axis between $x = a$ and $x = b$

Then

(A) $2\pi KL < S < 2\pi A$

(B) $S \leq 2\pi A < 2\pi KL$

(C) $2\pi A \leq S < 2\pi KL$

(D) $2\pi A < 2\pi KL < S$

Question 13 (JAM 2005 • Q13 • ID: JAM_2005_Q13)

MCQ

+6 M

Neg: -2.0

13. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(t) = t^2$ and let U be any non-empty open subset of $\mathbf{R}$.

Then

(A) $f(U)$ is open

(B) $f^{-1}(U)$ is open

(C) $f(U)$ is closed

(D) $f^{-1}(U)$ is closed

Question 14 (JAM 2005 • Q14 • ID: JAM_2005_Q14)

MCQ

+6 M

Neg: -2.0

14. Let $f: (-1, 1) \rightarrow \mathbf{R}$ be such that $f^{(n)}(x)$ exists and $|f^{(n)}(x)| \leq 1$ for every $n \geq 1$ and for every $x \in (-1, 1)$. Then f has a convergent power series expansion in a neighbourhood of

(A) every $x \in (-1, 1)$

(B) every $x \in \left(-\frac{1}{2}, 0\right)$ only

(C) no $x \in (-1, 1)$

(D) every $x \in \left(0, \frac{1}{2}\right)$ only

Question 15 (JAM 2005 • Q15 • ID: JAM_2005_Q15)

MCQ

+6 M

Neg: -2.0

15. Let $a > 1$ and $f, g, h : [-a, a] \longrightarrow \mathbf{R}$ be twice differentiable functions such that for some c with $0 < c < 1 < a$,

$$f(x) = 0 \text{ only for } x = -a, 0, a;$$

$$f'(x) = 0 = g'(x) \text{ only } x = -1, 0, 1;$$

$$g'(x) = 0 = h'(x) \text{ only for } x = -c, c.$$

The possible relations between f, g, h are

- (A) $f = g'$ and $h = f'$
- (B) $f' = g$ and $g' = h$
- (C) $f = -g'$ and $h' = g$
- (D) $f = -g'$ and $h' = f$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2005 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	C
2	MCQ	+6	A
3	MCQ	+6	B
4	MCQ	+6	D
5	MCQ	+6	D
6	MCQ	+6	C
7	MCQ	+6	D
8	MCQ	+6	D

Q. No.	Type	Marks	Official Key
9	MCQ	+6	C
10	MCQ	+6	A
11	MCQ	+6	A
12	MCQ	+6	C
13	MCQ	+6	B
14	MCQ	+6	A
15	MCQ	+6	B

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.